MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, GERMANIUM,

TYPE JAN-1N933

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a germanium electrically pulse formed junction (point contact) semiconductor diodes with a controlled conductance.

1.2 Ratings.

	V_r	V_R	$V_{R(rms)}$	i_f	I_o T_A = 25° C	I_o T_A = 75° C	$i_{f(surge)}$ 1 sec	P_T 1/ T_A = 25° C	Barometric pressure
	vdc	Vdc	Vac	ma	mAdc	mAdc	ma	mW	mm Hg
Minimum-	---	---	---	---	---	---	---	---	15
Maximum-	-100	-80	57	90	30	10	400	80	---

1/ Derate power 12.3 mW for each 10° increase in ambient temperature above 25°C.
Derate voltage 10 percent for each 10° C increase in ambient temperature above 25°C.
NON-OPERATING TEMPERATURE: -65° TO +90°C.
OPERATING TEMPERATURE: -65° TO +75°C.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-19491 - Semiconductor Devices, Preparation for Delivery of.
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

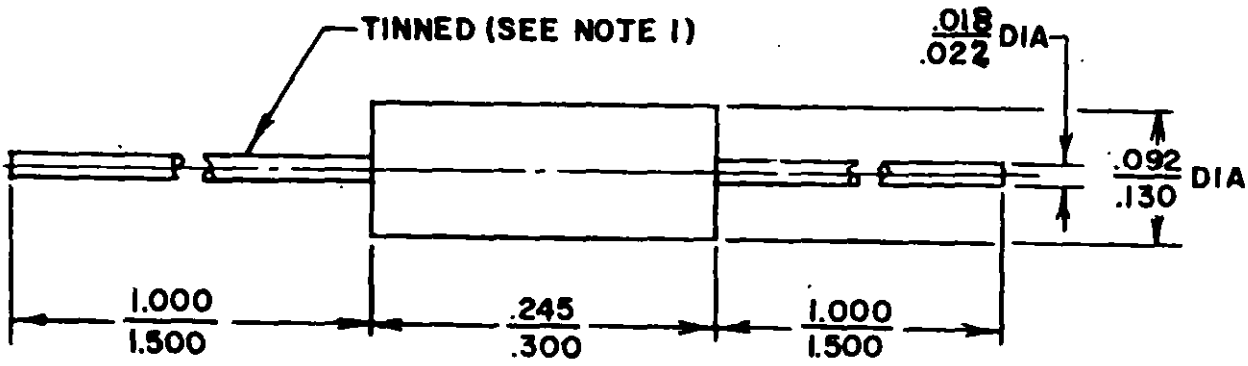
MILITARY
MIL-STD-750 - Test Methods for Semiconductor Devices.

DRAWING
ARMED SERVICES ELECTRO-STANDARDS AGENCY
256-JAN - Reverse Pulse Recovery Time Test and Calibration Procedure.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5960

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NOTE:

1. Gold plated leads may be substituted when specified in the contract or order, provided gold plated units conform to subgroup 5 of group B inspection. (See Table II.)

Figure 1. *Semiconductor device, diode, type JAN-1N933.*

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3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with Specification MIL-S-19500, and as specified herein.

3.2 Design and construction. The semiconductor diode shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.4 Marking. The following markings shall be placed on the device.

(a) Qualification-designating code.
(b) Type designation.
(c) Acceptance date and inspection lot identification.
(d) Polarity marking (a contrasting color band may be used to indicate cathode.)
(e) "JAN" prefix ("J" prefix may be used).

3.4.1 The manufacturer at his option may include the following markings:

(a) Country of origin.
(b) Manufacturer's identification.

3.4.2 All of the above markings except polarity shall be placed on the unit package.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with Specification MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.2.1 Qualification sample sizes and acceptance numbers. For purposes of qualification inspection, the LTPDs listed in this detail specification shall govern. The sample sizes and acceptance numbers shall be as listed in tables II and III of Specification MIL-S-19500.

Table I. Group A inspection.

Examination or test	Symbol	MIL-STD-750		Limits			LTPD	Min Rej No
		Method	Details	Min	Max	Units		
Subgroup 1 Visual and mechanical examination	---	2071		---	---	---	5	4
Subgroup 2 Forward current	I_F	4011	V_F = 1 Vdc	4	17	mAdc	5	4
Reverse current	I_R	4016	V_R = 10 Vdc	---	10	uAdc		
Reverse current	I_R	4016	V_R = 80 Vdc	---	80	uAdc		
Drift	ΔI_R	3015	V_R = 80 Vdc	---	5	uAdc		
Reverse current at peak reverse voltage.	I_R	4016	v_r = 100 vdc	---	500	uAdc		
Reverse recovery time to 0.5 mA (80 K) level.	t_{rr}		Drawing 256-JAN; Cond. A I_F = 5 mA; V_R = 40 V; R_L = 2 K	---	0.4	usec		
Reverse recovery peak pulse.	i_r		Drawing 256-JAN; I_F = 5 mA; V_R = 40 V; R_L = 2 K	---	1.1	mA		
High-temperature operation:	---		T_A = 75° C min	---	---	---		

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Table I. Group A inspection (cont'd).

Examination or Test	Symbol	MIL-STD-750		Limits			LTPD	Min Rej No
		Method	Details	Min	Max	Units		
Forward current	I_F	4011	V_F = 1 Vdc	5	21	mAdc		
Reverse current	I_R	4016	V_R = 10 Vdc	---	75	uAdc		
Reverse current	I_R	4016	V_R = 50 Vdc	---	250	uAdc		
Capacitance	C C	4001	V_R = 1.5 Vdc V_R = 9.0 Vdc	--- ---	1.0 0.7	uuf uuf		
Reverse current, average	I_{Ro}	4046	T_A = 75°C; Vac = 35 V; I_o = 10 mAdc	---	200	uAdc		

Table II. Group B inspection.

Examination or Test	Symbol	MIL-STD-750		Limits			LTPD	Min Rej No
		Method	Details	Min	Max	Units		
Subgroup 1 Physical dimensions	---	2066		---	---	---	---	---
Subgroup 2							10	5
Solderability	---	2026		---	---	---		
Temperature cycling	---	1051	T_A = -65°C max +90°C min	---	---	---		
Thermal shock (glass strain)	---	1056	Test cond A; Nonoperating; 90°±5°C 0°±5°C	---	---	---		
Moisture resistance	---	1021	Omit initial conditioning	---	---	---		
End points:								
Forward voltage	V_F	4011	I_F = 4 mAdc	---	1	Vdc		
Reverse current	I_R	4016	V_R = 10 Vdc	---	15	uAdc		
Reverse current	I_R I_R	4016	V_R = 80 Vdc V_R = 100 Vdc	--- ---	150 500	uAdc uAdc		
Subgroup 3 Shock	---	2016	Nonoperating; 500 G; 1.0 ms; 5 shocks ea in X_1 and Y_1 orientations				10	5
Vibration, variable frequency	---	2056	Nonoperating; 10 G	---	---	---		
Vibration fatigue	---	2046	Nonoperating; 10 G	---	---	---		
Constant acceleration	---	2006	Nonoperating; 10,000 G	---	---	---		
End points: (Same as for subgroup 2)	---			---	---	---		
Subgroup 4 Surge current	--- ---	4066 4066	Ten 1 sec Ten 1 sec surges; 1 surge/minute superimposed on I_o = 30 mA; i(surge) = 400 mA	---	---	---	10	5
End points: (Same as for subgroup 2)								

See footnote at end of table.

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Table II. Group B inspection (cont'd).

Examination or test	Symbol	MIL-STD-750		Limits			LTPD	Min Rej No
		Method	Details	Min	Max	Units		
Subgroup 5 1/								
Salt atmosphere (corrosion)	---	1041	96 hours	---	---	---	10	5
End points: (Same as for subgroup 2)	---	---		---	---	---		
Subgroup 6 1/								
Terminal strength (tension)	---	2036	Test cond. A; 2 lb; 1 minute	---	---	---	10	5
Terminal strength (lead fatigue)	---	2036	Test cond. E; 6 arcs	---	---	---		
End points: (Same as for subgroup 2)	---	---		---	---	---		
Subgroup 7								
Steady state operation life	--- ---	1026	Vac = 30 V; I_O = 30 mA	--- ---	--- ---	--- ---	=5.0%	4
End points: (Same as for subgroup 2)	---	---		---	---	---		
Subgroup 8								
High-temperature life (nonoperating)	---	1031	T_A = 90° C	---	---	---	=5.0%	4
End points: (Same as for subgroup 2)	---	---		---	---	---		

1/ Destructive tests.

Table III. Group C inspection.

Examination or test	Symbol	MIL-STD-750		Limits			LTPD	Min Rej No
		Method	Details	Min	Max	Units		
Barometric pressure, reduced (altitude operation). 1/	I_R	1001	Pressure = 15 mm Hg; V_R = 30 Vac; I_O = 15 mA; t = 1 minute, min	---	100	uAdc	10	5

1/ This test shall be conducted on the initial lot, and thereafter every 6 months.

4.3 Acceptance inspection. Acceptance inspection shall consist of groups A, B, and C, and inspection of preparation for delivery (see 4.4).

4.3.1 Acceptance procedure. When a second sample is chosen, the total sample shall be associated with the minimum rejection numbers specified in table I, II and III.

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4.3.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.3 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.4 Group C inspection. Group C inspection shall consist of the test specified in table III.

4.4 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with Specification MIL-S-19491 to verify conformance with requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Specification MIL-S-19491.

6. NOTES

6.1 The notes specified in Specification MIL-S-19500 are applicable to this specification.

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